

A Low-Power Delay-Optimized High-Speed Dynamic Comparator with Temperature Compensation

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Abstract—Dynamic comparators play a crucial role in modern communication systems as one of the major blocks in analog to digital converters (ADCs). In such cases, a dynamic comparator has significant impact on parameters such as conversion rate, power efficiency, and overall ADC accuracy. The existing dynamic comparator designs often have some drawbacks including inefficiency for extremely small differential input voltages, high delay variability due to temperature changes, and high-power dissipation under variable conditions. The above problems become especially important when the comparator is used in low-power and high-speed applications like wireless communication systems, edge computers, and mobile sensors. To overcome the above disadvantages, this paper suggests designing an optimized low-power temperature-compensated dynamic comparator with a cascode cross-coupled preamplifier that increases differential gain and provides good working performance under extremely small differential input voltages. At the same time, temperature-compensated feedback is introduced in the latch part of the dynamic comparator to minimize delay variability under thermal changes. The new comparator design integrates the advantages of the high-gain preamplifier scheme and the simple low-power latch topology to provide thermal stabilization and energy efficiency. This work used the comparator using 45-nm CMOS technology and simulated the behavior of the comparator based on different temperatures and voltages using the Cadence Virtuoso simulator. The outcomes obtained show that the architecture used here offers better thermal stability and low power consumption characteristics needed for ADC designs.

Index Terms—CMOS design, delay optimization, dynamic comparator, high-speed circuits, low power consumption, temperature compensation.

I. INTRODUCTION

The demand for high-speed analog-to-digital converters (ADCs) has been growing exponentially due to their widespread application in modern systems such as communication interfaces, medical sensors, and edge computing systems. In an ADC, the comparator is one of the core elements which plays an important role in deciding conversion speed, resolution, and efficiency [1]. Dynamic comparators are widely employed in low-power and high-speed applications due to their compact size and absence of static power consumption [2].

Various architectural approaches have been proposed to improve the performance of dynamic comparators. In [3], the double-tail comparator incorporates clock gating to improve speed and reduce power consumption. This technique, however, increases area and complexity which may limit its application in miniaturized systems. In [4], there is an optimization of the number of comparators within a flash ADC which helps in area and power reduction. The introduction of multiplexers, however, creates delay and lowers noise performance which is detrimental to high-speed applications. The strong-arm latch comparator in [5] provides good power saving, but suffers from PVT sensitivity, requiring a clean clock signal making it unreliable in noisy environments. Enhancing accuracy through techniques such as kickback noise reduction add design complexity and overhead [6]. The two-stage dynamic comparator in [7] claimed improvements in speed and precision.

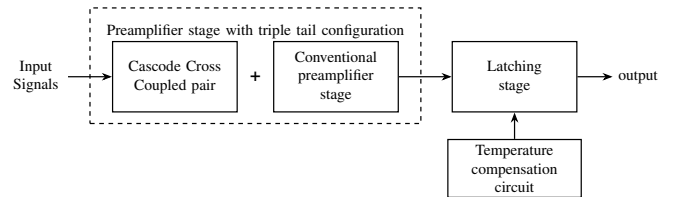


Fig. 1: Block diagram of the proposed comparator architecture.

Various level shifter designs strive to try and optimize voltage level compatibility in multi-Vdd systems. The Cascode Current Mirror-based Level Shifter (CCMLS) [8] does efficient conversion but doesn't operate well at low voltages. In the same fashion, the comparator in [9] was able to strike speed and power balance but also became very sensitive to noise which destroys low input differential stability. Improved strong-arm in [10] gains more but uses even more power. On the other hand, two-stage dynamic comparator in [11] suffers from non-compact scalable layout integration.

In [12], employing a transconductance-boosted latch aids in regeneration speed, but contributes no other form of power saving leading to a high dynamic power penalty under frequent switching. The Triple-Tail comparator in

[13] increases differential gain and common-mode rejection with a cascode cross-coupled preamplifier achieving fast decision making under small input voltage. However, the work does not propose any power optimization plan, resulting in power-critical designs.

In our previous work [14] incorporated a delay optimized dynamic comparator with temperature compensation for the latch stage to stabilize delay under thermal variation. Nevertheless, the compensation circuitry added on worsened overall power consumption, showing there is room for improvement from a power efficiency perspective. In addition, power and signal optimization are exhibited in new level shifters and sequential circuit design.

The Robust Body Biased Level Shifter (RBBLS) minimizes static power using body biasing and error correction but suffers from frequency limitation and poor scalability. The Rail to Rail low power D Flip-Flop is based on a current-mode logic structure and PMOS current mirrors to save power, but always-ON load devices are responsible for static power dissipation. The Modified Revised Wilson Current Mirror Level Shifter has an efficient subthreshold to supply voltage conversion, but high transistor count raises layout area and complexity. MCRLS design provides excellent voltage adaptability and body biasing support, but it encounters high delay as well as more power at lower voltages.

In contrast with all these drawbacks, we propose a hybrid dynamic comparator that integrates the high-gain preamplifier from the triple-tail architecture [13] with a simplified, low-power latch stage described in [14]. The architecture of the design can be seen in Fig. 1. The design employs cascode cross-coupling which provides high common mode rejection and additional gain, all while employing a latch design which is capable of minimizing power dissipation without sacrificing delay stability at different temperatures.

Therefore, the proposed design solves the most critical problems highlighted in [1]–[14] by circumventing gated and multiplexed comparators' area complexity and impeded metrics, overcoming the noise sensitivity and high static power of the level shifter-based approaches, and providing isothermal delay with low energy expenditure, which enabling the use of such a method for ADC devices with lower power consumption and enhanced heat resistance.

This paper is structured as follows: The Section II presents a survey of conventional comparators and the triple-tail comparator architecture. The Section III proposes a new comparator design and provides an operational analysis. Section IV discusses the simulation results alongside a performance analysis. Lastly, Section V discusses the conclusions that highlight the major insights and prospective applications.

II. REVIEW OF EXISTING WORK

A. Conventional Dynamic Comparator

A typical comparator with a dual-tail arrangement is made up of a latch stage and an input, each of which has a tail transistor. We can more effectively optimize the trade-off

between speed, offset, and energy consumption by using independent tail transistors to control the tail currents. As seen in Fig. 2, it makes use of 14 transistors, four of which are set up as two linked inverters. The inversion property of inverters allows the latch stage to both hold and amplify the binary outputs, 0 and 1, at the appropriate output points.

Every conventional dynamic comparator function in two separate stages.

- 1) Reset Phase
- 2) Evaluation phase.

The clock is set to a low value at the beginning of the reset phase, which causes the two PMOS devices M3 and M4 to conduct and charges fp and fn to V_{dd} . As a result, NMOS transistors MR1 and MR2 receive a high enough voltage to activate. OutN and OutP are then discharged to GND, where the output nodes are referred to as reset. To guarantee correct circuit initialization prior to the evaluation phase, MT1 and MT2 typically remain disabled at this point.

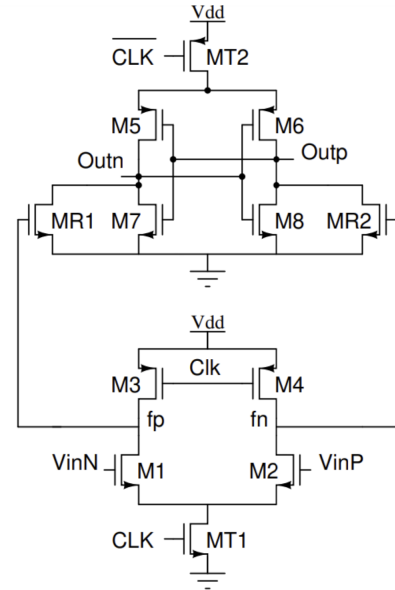


Fig. 2: Schematic of conventional dynamic comparator.

When the clock is set high at the beginning of the evaluation phase, the PMOS transistors M3 and M4 are deactivated and the tail transistors MT1 and MT2 are activated. As a result, depending on the corresponding input voltage levels, V_{inN} and V_{inP} , the output points that were grounded during the reset phase start to charge or discharge at varying rates. dp discharges more quickly than dn when $V_{inN} > V_{inP}$, causing MR1 to shut off sooner. OutP begins discharging through MR2, which supplies a low enough voltage to M5's gate, causing OutN to be charged through M5 and MT2. Likewise, the functionality is maintained when $V_{inN} < V_{inP}$.

The sensitivity of a conventional double-tail dynamic comparator to tiny input differential voltages is one of its main drawbacks. Due to the restricted pre-amplifier gain, the latch may occasionally have issues sensing signals when the input voltages are almost identical. To maximize the differential

gain, a cascode cross-coupled arrangement is incorporated into the pre-amplifier, as explained in [11].

B. Existing High-Speed Dynamic Comparator with Triple Tail Configuration

The high-speed Triple-Tail comparator architecture is constructed using a modified double-tail structure, in which the preamplifier stage is enhanced by a cascode NMOS cross-coupled pair to increase common-mode in-sensitivity and improve differential gain [13]. The detailed transistor level implementation is shown in Fig. 3.

During the reset interval ($CLK = 0$), tail devices MT1, MT2, and MT3 are turned off, while the cascode devices M3 and M4 are also inactive. At the same time, the transistors MS1 to MS4 are in the on state and are charging the intermediate nodes (drains of M1 to M4) to Vdd. Also, the reset transistors MR1 and MR2 are actively discharging the outputs OutN and OutP, which guarantees the proper setting of the latch stage before the evaluation is performed.

When in evaluation phase, the clock signal oscillates to a high state ($CLK = 1$), turning on MT1, MT2, and MT3. The switching transistors become OFF, isolating the nodes from Vdd. From the start, the XCP created by M3–M6 is in off state since M3 and M4 is zeroed out due to V_{GS} . With a differential input, in a Next Step Cycle, the voltages at one of the input nodes falls quickly, resulting in a small voltage difference at the intermediate nodes, which starts to form. This difference causes the gate voltages of M3 and M4 to boost, surpassing the threshold which turns them ON.

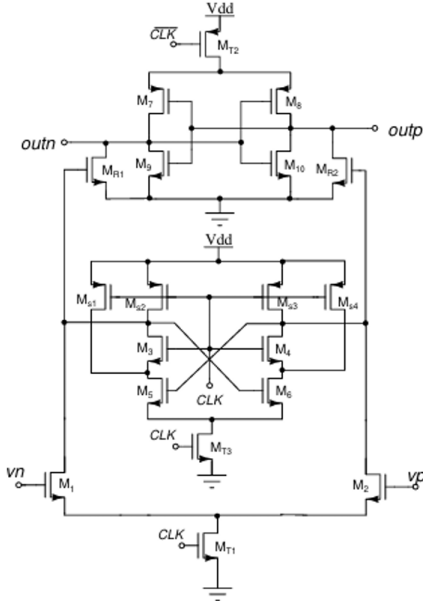


Fig. 3: Schematic transistor-level circuit of triple-tail common-mode insensitive dynamic comparator developed to be used in high-speed analog in-memory computing applications.

After activation of the cascode cross-coupled pair (XCP), this works within the saturation region and amplifies the voltage difference generated at the preamplifier outputs. Such

a high degree of differential reduces the latch regeneration time, therefore, fastening the decision-making process of the output nodes.

This study focuses on mitigating the delay dependency on the temperature variation while retaining the high-speed and high-resolution benefits that come with the cascode configuration at the input stage. To further reduce the delay, a small initial voltage is given to the output. The input stage of the presented comparator operates similarly to that of [11] both in the reset phase and the evaluation phase.

However, the functionality of the proposed comparator varies when it comes to the latch stage. Unlike conventional comparators, where the output terminals are discharged towards the ground, the reset voltage V_{set} of the proposed comparator depends on the reference voltage V_{ref} and R_1 . V_{ref} is given from the external power, and R_1 is the resistor invariant to the temperature.

The triple-tail configuration thus allows for rapid high-speed comparison even with very minimal differential inputs achieving 49 ps and 36.75 ps delays for 1 mV and 10 mV input difference, respectively. However, this architecture, while fast, enhances common mode functionality at the expense of adaptable temperature compensation, power saving features, and added lower power conditions in varying environmental conditions.

III. PROPOSED LOW-POWER TEMPERATURE-STABLE DYNAMIC COMPARATOR

A. Circuit Overview

The comparator uses a modified triple-tail configuration comprising a differential input pair [13] to the delay optimized high-speed dynamic comparator [14], a cross coupled latch, and a control tail as shown in Fig. 4. It employs a tri-path current steering scheme as each of the input, latch, and control stages have separate current sources which improves speed and isolation over traditional approaches. The tri-path current steering mechanism provides each of the input, preamplifier and latch stages to be independently biased. Architecture enables optimization of delays without a corresponding power consumption increment.

The dynamic comparator's propagation delay is measured by moment the voltage difference at the two output terminals reaches half of Vdd. This includes t_{amp} , t_0 , and t_{latch} . In this work, we are focusing on advancing the circuitry so that t_0 is reduced since that will reduce the overall delay. The value of t_0 corresponds to the time duration defined in the caption, which in our case is the time the output signals are transitioning from V_o to V_{th} of transistors M5 and M6. Fig. 5 illustrates. To determine this the following formula is utilized:

$$t_0 = \frac{(V_{tn,5} - V_0) \cdot C_{out}}{I_{m11}/2} \quad (1)$$

The propagation delay for the comparator can be expressed by the following equation:

$$t_{delay} = t_0 + t_{amp} + t_{latch} \quad (2)$$

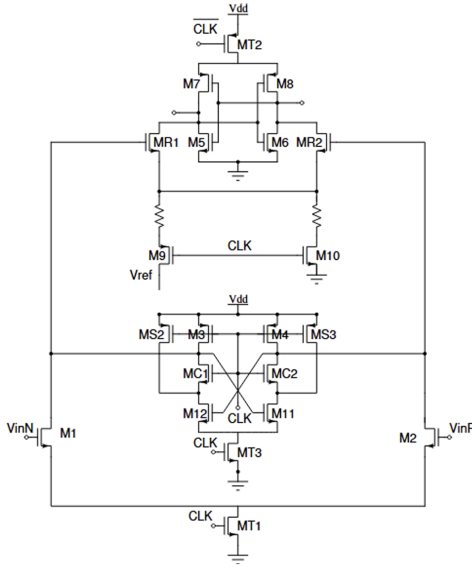


Fig. 4: Schematic of proposed Low-Power Delay-Optimized Dynamic Comparator.

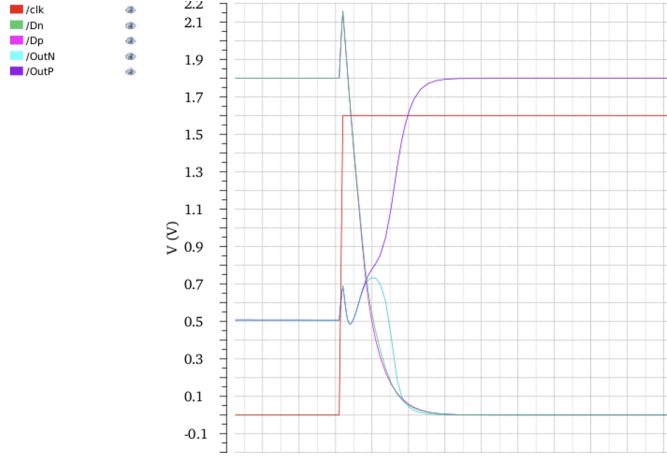


Fig. 5: Waveform for the proposed dynamic comparator.

From (1) and (2), it is obvious that t_{delay} can be reduced by reducing t_o . Set V_{set} to V_o during the entire evaluation phase. As we know, the energy consumption in a conventional comparator increases because of temperature because of slow transistor switching. To help fix this issue, the proposed circuit incorporates a temperature-dependent resistor R_2 that resides in the latch part of the circuit. The overriding temperature coefficient ensures that R_2 is positively bounded such that increasing the temperature leads to an increase in R_2 , which increases the voltage at the V_o node. Since the output rises and the fall time is faster, this will help to reduce t_0 in the way served on Fig. 5.

B. Circuit Operation

Reset Phase (When CLK = 0):

When the CLK = 0, Tail transistors, MT1, MT2, and MT3 are turned off and the comparator is put in the non-conducting state. A zero static power is achieved when the circuit is in

reset phase. The transistors MC1, MC2 also turns OFF and the transistors MS1, MS2, MS3, MS4 turns ON, which charges the drains of VinN, VinP and the nodes Dn and Dp. The gates of MR1 and MR2 get sufficient high voltage to turn ON, which results in the output nodes OutP and OutN resetting to V_{set} .

Evaluation Phase (When CLK = 1):

When CLK turns active (CLK = 1), all tails transistors MT1, MT2 and MT3 turn ON making the circuit active. MS1, MS2, MS3 and MS4 turn OFF. Suppose $V_{inN} > V_{inP}$, which makes M1 more active than M2. Thus, Dp discharges faster than Dn, providing potential difference across the pre-amplifier stage. M4 gets sufficient low voltage to turn OFF faster than M3. DP keeps on discharging and Dn stops discharging at higher voltage. This creates more differential input at the latch stage, which in turn increases the accuracy and the speed of the comparator.

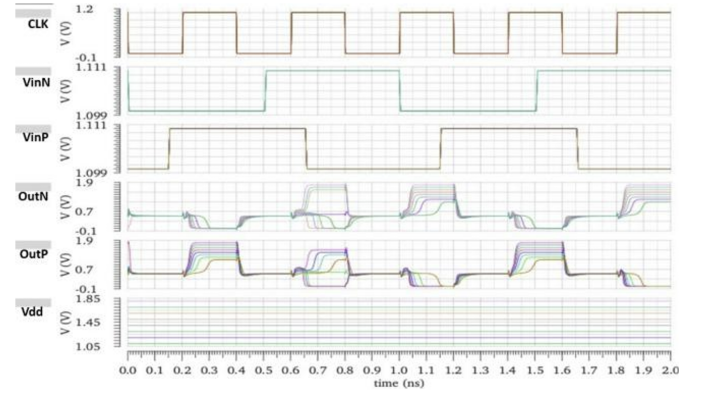


Fig. 6: PVT analysis for constant temperature.

PVT analysis is performed for the proposed circuit using ADE XL in Cadence Virtuoso by keeping the supply voltage constant at 1.1 V while varying the temperature from -40°C to 120°C (Fig. 7) and keeping the temperature constant at 20°C while changing the supply voltage from 1.1 V to 1.8 V (Fig. 8).

IV. RESULTS AND DISCUSSION

The same device parameters were used in all circuit implementations so as to have a level playing field in the evaluation of performance. The analysis were carried out using a 45 nm CMOS process with a supply voltage of 1.1 V. Fig. 6 illustrates the transient response of the proposed comparator with a 10 mV differential input. During the low clock phase (reset phase), the two outputs OutN and OutP are set to V_{set} . Upon CLK transition to 1, if $V_{inP} > V_{inN}$, OutP pulls up and OutN drops, thus OutN decreases towards GND. Conversely, when $V_{inN} > V_{inP}$, OutN rises to Vdd and OutP falls to GND. The transient response confirms correct operation of proposed architecture nodes settle to complementary logic states in the following clock transition. The early amplification reduces metastability probability and improves delay stability across operating conditions.

The detailed analysis of power consumption [14] is alongside the proposed circuit at varying supply voltages from

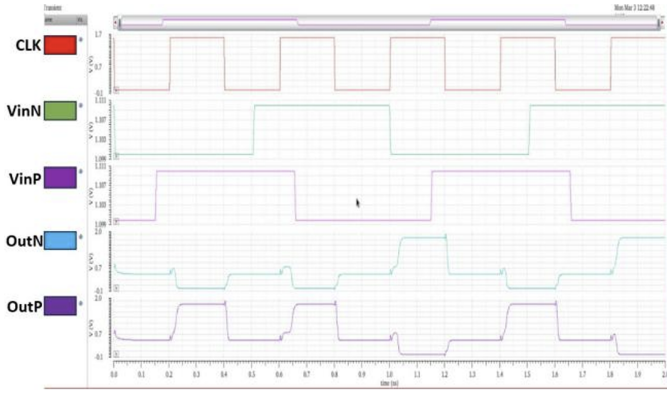


Fig. 7: Transient analysis for the designed comparator.

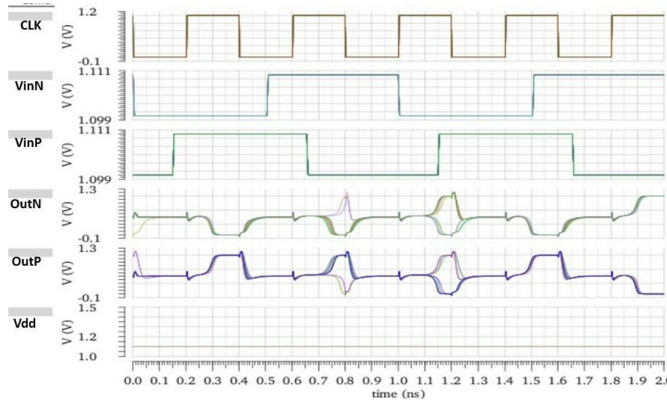


Fig. 8: PVT analysis for constant voltage.

1.8 V to 0.5 V, which proves that the proposed comparator consumes less power compared to the others. In the same way, a comprehensive delay analysis has been performed on the conventional comparator along with [14] and the proposed comparator shown in Fig. 10 with their slopes.

The slope result obtained from the simulations clearly shows that the delay time of the designed comparator is less affected by changes in temperature than the traditional comparator and [14]. The performance analysis of the current design relative to other existing designs is shown in Table I below, wherein all circuit structures were analyzed using similar parameters: 45 nm CMOS process, 1.8 V power supply, and 2.5 GHz clock speed.

Even though the suggested comparator has greater delay (67.78–86.25 ps) than [11], [13], and [14] comparators, this is a conscious decision made during design. This design aims at lower power dissipation (56.22% lower than that of [14]) which is 7.90 μ W against 19.13 μ W of [14], and therefore it can be used in applications with energy constraints like SAR ADCs with low speed and high accuracy where delay is not a primary consideration. Slope of the delay vs. temperature graph (0.115 ps/ $^{\circ}$ C) proves that temperature stability is ensured.

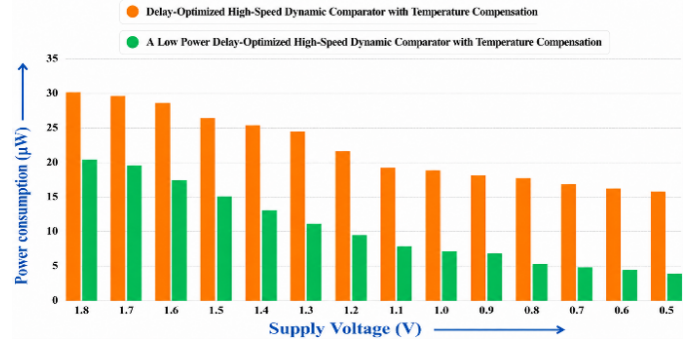


Fig. 9: Power consumption analysis for proposed circuit and [14] at different supply voltages.

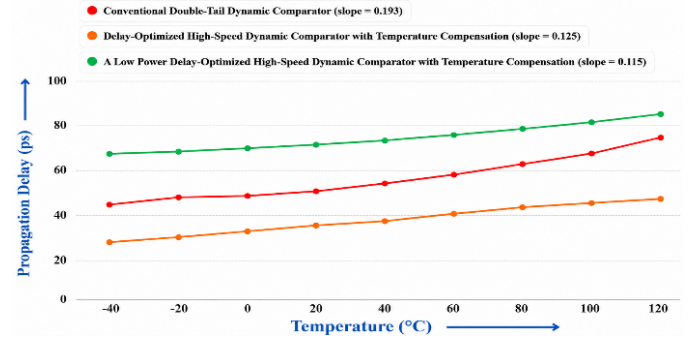


Fig. 10: Delay analysis for conventional circuit, [14] and the proposed comparator.

V. CONCLUSION

Implementing a comparator that has better temperature stability characteristics along with low power consumption by utilizing a novel preamplifier stage. Results show that the proposed comparator achieves a 56.22% reduction in power consumption and delay variation by 8% and 40.41% when compared to the conventional comparator. The ability to consume very little power and also have a temperature insensitive delay makes it an ideal candidate for use in flash and successive approximation register (SAR) ADCs that operate in thermal environments where the temperature keeps varying.

TABLE I: Comparison of Dynamic Comparators Under Various Temperature Conditions

Parameters	Conventional	[11]	[13]	[14]	Proposed
Tech. Used (nm)	45	45	45	45	45
Supply V (V)	1.8	1.8	1.8	1.8	1.8
Clock Freq. (GHz)	2.5	2.5	2.5	2.5	2.5
Power Cons. (μ W)	3.22	7.36	7.56	19.13	7.90
Prop. Delay (ps)					
−40 $^{\circ}$ C	45.03	21.68	26.64	28.13	67.78
−20 $^{\circ}$ C	48.72	23.17	28.27	30.62	68.69
0 $^{\circ}$ C	49.26	24.79	29.98	33.72	70.14
20 $^{\circ}$ C	51.72	26.50	31.84	36.62	71.83
40 $^{\circ}$ C	55.83	28.31	33.78	38.35	73.86
60 $^{\circ}$ C	59.27	30.29	35.87	41.78	76.54
80 $^{\circ}$ C	64.27	32.33	38.03	44.72	79.45
100 $^{\circ}$ C	69.28	34.46	40.31	46.62	82.64
120 $^{\circ}$ C	75.91	36.71	42.73	48.23	86.25
No. of Transistors	14	21	21	24	23
PDP (fJ)	175	244	297	792	665

ACKNOWLEDGMENT

The authors acknowledge the support of the Department of Electronics and Communication Engineering, Amrita School of Engineering, Bengaluru, Amrita Vishwa Vidyapeetham for providing the facilities to carry out the work.

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